Eskom	Scope of Work	Eskom Real Estate

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1. Introduction

Eskom Real Estate (ERE) currently manages access at Eskom corporate sites (Eskom Megawatt Park, Eskom Academy of Learning and Eskom Research and Innovation Centre). Security and safety of all people and property entering and exiting the sites are of paramount important to Eskom.

Multi-layer security measures are important to be in place and fully always functioning. The security measures must meet the safety and security requirements as per the regulations of the country and world standard practice.

Visitor management is very important for the organisation to have a seamless experience of access to premises for visitors to be able to digitally account for all the visitors in the premises. The personal information stored shall be secured from third party interference. The information stored shall be credible and POPIA compliance be adhered to.

Technology allows for digitalisation of information, quick retrieval of information, and the integration of artificial intelligence for visitor management.

2. Background

Eskom Real Estate (ERE) currently manages visitor access at Eskom corporate sites (Eskom Megawatt Park, Eskom Academy of Learning and Eskom Research and Innovation Centre) through manual paper-based check in/out process, which presents several operational and security challenges. Manual recording results in slow processing times, incomplete or inaccurate visitor information, and limited traceability when historical data is required for audits or investigations. This approach also places unnecessary administrative pressure on security personnel, reducing their ability to focus on other core security responsibilities.

Manual system tends to be susceptible for challenges due to manual filling and storage of documents. Eskom's governance, compliance, and safety requirements, including POPIA, access control standards, and emergency evacuation procedures has technologically advanced and created a misalignment with manual visitor registration or booking. The inability to generate real time visitor data, verify identities efficiently, or produce accurate occupancy reports creates risks for both visitors and employees.

Over time, existing security technologies have deteriorated due to technological advancements, inadequate maintenance, and an increasingly complex crime risk landscape. These factors have contributed to significant vulnerabilities, exposing the campus to potential operational and asset losses.

To address these challenges, a comprehensive review and enhancement of physical security infrastructure is essential. The objective is to mitigate current and emerging threats through the deployment of advanced, AI and IoT driven security technologies.

3. Purpose

Eskom invites potential suppliers to submit proposals for a turnkey solution covering the design, supply, installation, commissioning, and maintenance of the visitor management system and Licence Plate Recognition (LPR) at Eskom corporate sites (ERIC, MWP and EAL). This contract shall be implemented under a technology as a service model, specifically on an Outcome Based Contracting (OBC) basis for a duration of three years.

This document serves as an overview of the scope for visitor management system and stipulates the guidelines of the required deliverables or requirements.

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It also provides detailed technical specifications and minimum performance requirements for visitor management system to be installed at entrances and reception areas of the sites.

The system must meet all operational, safety, and integration requirements suitable for critical infrastructure security environments.

The visitor management system will be used to manage the access rights of Corporate Sites employees, visitors and contractors in and out of different areas at all the sites.

The system will also be used to grant and limit access permissions in and out of secure and non-secure areas.

The system should support a tiered architecture, which will allow monitoring of the sites both locally and remotely. This architecture comprises field devices (biometric, QR codes, card readers, facial recognition) at the site level and system management servers at the remote security nerve centre.

The system should be modern and fit for a smart initiative.

4. User requirements baseline

- Easy to use system
- Off-site visitor self-registration system
- QR use system
- Face recognition system (picture on self-pre-registration)
- Reject and flag expired drivers licence registration
- Reject and flag expired registration number plate
- Vehicle registration plate recognition
- Only one option selected to be used at a time (QR use or Face recognition)
- System to allow for mass QR invitation
- Be able to integrate or interface with employee management system
- Host approval then security management approval

5. Supporting clauses

5.1 Scope

The scope of VMS work will include the following but not limited:

- Supply and setup of all required hardware (such as kiosks, tablets, scanners, AI and IoT devices).
- Integrate with existing access control systems, network infrastructure, and user identity platforms.
- The full installation, configuration, and commissioning of the Visitor Management System (VMS) across designated Eskom Real Estate corporate sites.
- Deploy the VMS software, configure workflows, dashboards, and security settings.

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- Enable smart features such as visitor pre-registration, host notifications, QR/ ePass issuance, and real-time monitoring.
- The testing of the system functionality, data protection controls, and performance across all user touchpoints.
- Full system documentation and handover.
- Validating system readiness, ensuring full operational functionality, and providing technical support during the initial rollout period.
- Produce basic and detailed designs for the VMS. The detailed design must include detailed process flow of the VMS from invitation of the visitor to the visitor leaving site.
- Present the designs to the security team for acceptance. Provide services as per 240-170000723.
- Installation, configuration and commissioning of the VMS in totality on site as per Eskom Standard (240-102220945).
- Integration of the VMS System into an Integrated Security System (240-170000096) to interface with the PSIM system.
- Create a graphical user interface (GUI) and behaviour models for the site.

Must incorporate, integrate and interface the multiple access point areas including:

MWP:

- 7 x Fixed and 4 X handheld scanners for parking, both entrances (East and West).
- 12 x Kiosks for registrations or checking in at building access control points.
- 8 x high resolution vehicle registration plate recognition cameras for both entrances (East and West).

EAL

- 3 x Fixed and 4 X handheld scanners for main and emergency gates.
- 11 x Kiosks for registrations or checking in at building access control points.
- 6 x high resolution vehicle registration plate recognition cameras for both entrances.

ERIC

- 3 x Fixed and 3 X handheld scanners for the gate.
- 08 x Kiosks for registration or checking in at the building.
- 4 x high resolution vehicle registration plate recognition cameras at the entrance.

- Local servers should integrate to the main off-site server as a backup.
- Server's information retention period must be a minimum of six years.
- The subsystems must interface with a Physical Security Information Management (PSIM) system, including the necessary IT and OT infrastructure.

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- Bidders are required to use the accompanying technical specifications/ standards when preparing their proposals. Proposals must include a technology roadmap or project plan.
- A three-year support and maintain of the systems to ensure sustainability, continuous improvement, and alignment with the evolving risk landscape and technological improvements.
- Visitors can pre-register through an online portal or a mobile app before their visit. For permission, the system gathers identity documents, pictures, personal information, the reason and duration for the visit.
- Set up a self-service kiosk or tablets at the appropriate location for visitors check in or check out.
- For security reasons, guests will need to take a picture, digitally sign in, or scan their identification card.
- The ability to give and deny access to authorised areas based on visitor data is one of the features of integration with the present access control system that ensures compatibility.
- Identification and verification: identity verification techniques, including QR code scanning, ID scanning, driver's license verification, and biometric choices (such as fingerprint or face recognition).
- Following a successful check-in, visitors are given a physical or digital badge that contains their name, photo, the locations they are permitted to enter, and the host's information.
- Host notification: setting up to notify hosts in real-time when guests arrive (visitors can only proceed when the host collects them from reception).
- Integration with access control systems, including key cards and biometric readers. The host should tag his or her card on a dedicated reader when fetching the visitor to automatically allow the visitor to use his or card/face/finger/ID for access at main reception entrance door or turnstile only.
- For compliance and audit trails visitor's movements from entrance to last site exit point must be recorded and stored.
- System should reject a visit application request on an expired drivers' licence and passports.
- Declares the incoming personal property or property that is intended to go out on the same day (for example laptop).
- Approval should be able to send the notification to the visitor about the house rules, induction and address or site GPS coordinates.

N.B. The systems specified represent baseline technology and do not preclude the tenderer from proposing more advanced alternative solutions.

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5.2 Applicability

This document shall apply to Eskom Corporate sites which are Eskom Academy of Learning (EAL), Eskom Megawatt Park (MWP) and Eskom Research and Innovation Centre (ERIC).

5.3 Effective date

This document will be effective from the date of authorisation.

5.4 Normative/Informative references

Parties using this document shall apply the most recent edition of the documents listed in the following paragraphs.

5.4.1 Normative

[1] ISO 9001 Quality Management Systems

[2] 240-102220945 Specification for Integrated Access Control System for Eskom sites

[3] 240-91190304 Specification for CCTV Surveillance with Intruder Detection

[4] 240-86738968 Specification for Integrated Security Alarm System for Protection of Eskom Installations and its Subsidiaries

[5] 240-170000096 Physical Security Integration Standard

[6] 240-170000257 Technical Evaluation Criteria for the Integrated Security System

[7] 240-60725641 Specification for Standard (19-inch) Equipment Cabinets

[8] DEM2412993 & 2425114 LAD (Logical Architecture Definition) PAC (Physical Application Component) for Physical Security Information Management System (PSIM)

[9] 240-170000723 Generic Technical Requirements for Physical Security Technologies Contracts

[10] 32-373: Information Security – IT and OT Third Party and Remote Access Standard

[11] 240-55410927 Cyber security standard for Operational Technology

[12] 0.54-393 Earthing Standards

5.4.2 Informative

N/A

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5.5 Definitions

5.5.1 General

Definition	Explanation
Tender	Refers to an open or closed competitive request for quotations/prices against a clearly defined scope/specification.
Integrated Access Control System	It is an electronic system that aims to collaborate and align efforts across the logical and physical security domains to standardise access control within Eskom Academy of Learning.
Control Centre	Where alarms and CCTV footage are monitored and needed response/s initiated from. The alarms and CCTV footage can be aggregated to a National Security Control Centre that can initiate requisite actions from a national perspective.
Contractor	The term will be used interchangeably with tenderer, service provider, supplier or bidder.

5.5.2 Abbreviations

Abbreviation	Explanation
EAL	Eskom Academy of Learning
AC	Alternating Current
MWP	Eskom Megawatt Park
DC	Direct Current
DVR/NVR	Digital Video Recorder/Network Video Recorder
GUI	Graphical User Interface
IoT	Internet of Things
IPSS	Integrated Physical Security System
IACS	Integrated Access Control System
LAN	Local Area Network
PSIM	Physical Security Information Management
UPS	Uninterrupted Power Supply
ERIC	Eskom Research and Innovation Centre
PSIRA	Private Security Industry Regulatory Authority
WAN	Wide Area Network
PIR sensor	Passive Infra-Red sensor
SSP	Security Solutions Physical
AI	Artificial Intelligent

5.6 Roles and responsibilities

The contractor should specify the roles and responsibilities of each of their team members. Roles shall be as outlined in 240-170000086.

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5.7 Process for monitoring

Not applicable

5.8 Related/ Supporting documents

Not applicable

6. Biometrics, QR codes and card readers

The main access points to the premises should use either a fingerprint, a SA identity card, a SA valid driver's licence, face recognition and a SA valid passport:

- Biometrics and card reader shall comply with SANS 2220-2-5 and SANS 2220-2-3, respectively.
- Readers (i.e. access card (Electronic or physical), biometric, face recognition, QR, ID) should be installed at main access points.

7. License plate recognition (LPR) cameras:

- High-resolution cameras capable of capturing license plates in various lighting and weather conditions.
- Integration with visitor management control systems to automate vehicle entry/ exit.
- Integrate the driver's licence to eNatis for real time alerts of expired licence, flag the expired licence and reject the application.
- Integrate the vehicle registration number to eNatis for real-time alerts of the expired vehicle disc, flag the registration number and reject the application.

8. Power supply

a) All system servers shall be housed in 19-inch equipment cabinets as specified in the Eskom Standard 240-60725641. This specification covers the earthing requirements in the cabinet as well.

b) Power shall be distributed through the panel to isolate the supply of the subsystems using appropriately sized MCBs.

9. Earthing and Bonding

The contractor shall conduct a comprehensive inspection of the earthing and bonding system.

The following requirements apply:

- The contractor shall inspect, test, and verify the earthing system for compliance with Eskom standards and applicable SANS electrical safety requirements. For:

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  - Personnel safety.
  - Protection of electronic equipment.
  - Compliance with electrical standards.
- Any faulty, damaged, high-resistance, corroded, or non-compliant earthing points shall be corrected, replaced, or reinstalled to meet specification.
- Measurements of earth resistance must be performed using calibrated instruments, and all results must be recorded and submitted.
- The contractor shall submit a detailed Earthing Compliance Report, including:
  - Test results
  - Corrective actions taken
  - Location maps of earth spikes and bonding points
  - Certification of compliance

10. As-Built drawings and documentations

The contractor shall prepare as-built drawings of their completed installation – including conduit, cables and fibre routes.

The Operator's Manuals must be compiled and contain enough detailed information to enable a suitably qualified technician to control and operate the full installation without any training from the contractor. The Operator's Manuals must be a separate set of documents from the Maintenance Manuals.

11. Training

All training aids and course notes necessary to conduct effective operational and maintenance training shall be supplied by the contractor.

The training venue will be made available on-site by Eskom.

Training shall be adequate to ensure that the groups trained are competent in the operation of systems, adequately trained to carry out ongoing training.

Trained groups should be fully aware of the location of all equipment installed within their areas of responsibility.

End users should know and understand how to use the system effectively.

12. Maintenance and Support

The contractor will be fully responsible for the maintenance and support of the installed systems throughout the 36 months period (preventive maintenance, scheduled inspections, system testing, and servicing) to ensure optimal performance and reduce the likelihood of failures.

All equipment shall carry a warranty of three years.

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The contractor must provide corrective maintenance by addressing system faults and restoring functionality within agreed timeframes.

The contractor must implement system upgrades as needed, incorporating relevant technology updates and improvements to maintain the system's effectiveness.

13. Site tests and commissioning

The testing of the system shall be done in the presence and to the satisfaction of an authorised Eskom representative.

The contractor shall do his/ her complete commissioning tests before the actual first take-over tests are done. This is to satisfy himself/ herself that everything is working and is in accordance with the required specification and scope.

14. Cyber Security Requirements

a. The Tender shall have a valid Information Security Standard (ISO) 27001 certificate**.**

b. The Tender shall have a valid International Organization Standardization (ISO) 15408 certificate.

c. The Tender shall have a valid International Organization for Standardization (ISO) 27034 certificate.

d. The system shall be able to integrate with existing Eskom's identity providers (IdP's) such as Microsoft (MS) Active Directory (AD), Entra Identity (ID) and shall support Security Assertion Markup Language (SAML) 2.0, Open Authorization (OAuth) 2.0 and OAuth 2.1 to enable Multi-Factor Authentication (MFA) and Single Sign-On (SSO).

e. The System shall employ Role-based access control (RBAC) mechanism.

f. The System shall be able to onboard privileged accounts onto Privileged Access Management (PAM) tools, such as CyberArk but not limited to onboard privileged accounts to securely manage, rotate passwords, monitor, record sessions, control, enforce access to privileged accounts, auditing and reporting capabilities.

g. Data at rest to be encrypted using at minimum Advanced Encryption Standard (AES)-256 and in transit (or in motion) using at minimum Transport Layer Security (TLS) 1.3 or later version.

h. Data shall be secured at minimum using a Secure Hash Algorithm (SHA)-256 for data integrity, securing transactions, and messages.

i. Audit trails, logs, user administration and user activity logs shall be enabled, encrypted, and securely kept with limited access to administrators.

j. Sensitive information such as personal identifiable information (PII) data in non-production environments shall be masked.

k. Incremental daily back-ups shall be done, encrypted, and securely kept offsite only for all critical systems.

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l. Real-time data synchronization or data replication to a secondary or disaster recovery (DR) site, located in different regions, shall be employed only for critical systems.

m. Disaster Recovery Plan (DRP) shall be defined, annually tested, and such DRP test results shall be shared with the Eskom Cybersecurity team only for critical systems

n. A back-up Restore Plan and Procedure shall be defined, annually tested, and such test results shall be shared with Eskom's Cyber Security team.

o. Patch Management Process shall be defined. The software updates and patches shall be tested in a non-production environment before being deployed into the production (PROD) environment.

p. The database shall be placed within Eskom corporate Local Area Network (LAN)/Business Information Network (BIN) network (if hosted on premise) and partner private network (If hosted in the cloud) behind the perimeter firewall.

q. Database Security Management tools shall be employed to provide regulatory compliance, encryption, key management, granular access controls, flexible data masking, comprehensive activity monitoring, and sophisticated auditing capabilities.

r. The Service shall support the prevailing enterprise services bus (ESB), application programmable interfaces (API's) and Integration Platform as a Service (iPaaS) platforms for security, logging and monitoring for both on-prem, hybrid-cloud and multi-cloud environments.

s. The on-premises System and Cloud Service shall be able to integrate with Security Information and Event Management (SIEM) standard technologies such as Syslog, Windows event logging, Simple Network Management Protocol (SNMP) and Application Programming Interface (API) but not limited to these technologies listed.

15. Warranty and Support

The contractor shall provide a three years (36 months) warranty for all installed systems from the date of final acceptance, covering defects in materials, workmanship, installation, and system configuration.

During the warranty period, the contractor shall, at no additional cost, rectify all failures attributable to such defects within the agreed response times, including repair or replacement of faulty components and limited preventive inspections solely for defect detection. Where required to maintain system operability, the contractor shall implement manufacturer-issued patches, firmware updates, or compatibility adjustments directly related to warranty compliance.

The contractor shall also provide limited refresher training and knowledge transfer to nominated stakeholders to ensure correct system operation and a smooth transition upon expiry of the warranty period, excluding damage arising from misuse, unauthorised modifications, third-party interference, or conditions outside specified operating parameters.

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16. Scalability and Future Readiness

- Be modular and scalable to support future expansion in screen size, resolution, number of sources, and integrated systems without major redesign.
- Support evolving technologies, including AI-driven analytics, advanced dashboards, and additional remote site integration.

17. Implementation timelines

Task	Duration	Milestones
System Design and Approval		Blueprint Finalisation
Equipment Procurement		Vendor Selection
Installation and Configuration		Deployment and Testing
AI & IoT Integration		Data Synchronisation
Training and Handover		Project Completion

Table: Implementation Timelines

18. Acceptance

This document has been seen and accepted by:

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19. Revisions

Date	Rev.	Compiler	Remarks

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20. Development Team

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21. Acknowledgements

N/A

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